Domain-Adversarial Generative and Dual-Feature Representation Discriminative Network for Hyperspectral Image Domain Generalization

Minghui Chu, *Graduate Student Member, IEEE*, Xiaodong Yu, Hongbin Dong, and Shuying Zang

***Abstract*— Traditional training models often experience significant performance drops on test sets when the training and testing data distributions differ. To address this domain shift problem, we propose D3Net, a domain generalization (DG) classification network for hyperspectral images (HSIs) based on generative adversarial networks (GANs). Specifically, D3Net consists mainly of a generator and a discriminator. The generator extracts domain-invariant information from the source domain to generate data with core classification features, while the discriminator employs a dual-confidence model to enhance the capture of domain-invariant features. Through adversarial iterations, the model is able to adapt to the domain shift effects of unknown data. Unlike existing DG methods that rely on random perturbations for data augmentation, D3Net utilizes learnable convolutional neural networks (CNNs) rather than randomization to enhance the model's learning capability. We conducted cross-scene classification experiments on datasets from Houston, Pavia, and Indiana, and the results demonstrate the effectiveness of our approach. The code for D3Net is available at: https://github.com/gmsjzyq123/D3Net.**

***Index Terms*— Contrastive learning, domain generalization (DG), generative adversarial network (GAN), hyperspectral image (HSI) classification.**

I. Introduction

With the rapid development of deep learning, significant progress has been made in visual tasks for RGB images [1], [2], [3]. Against this background, remote sensing tasks have gradually become a research hotspot. Remote sensing refers to the acquisition of target data using sensors without direct contact with the target, and obtaining useful information about detected targets, regions, and phenomena through data analysis. This has led to extensive research in a range of remote sensing tasks, such as hyperspectral image (HSI) classification, super-resolution, spectral unmixing, remote sensing image enhancement, and remote sensing image retrieval [4], [5], [6], [7]. Among these, hyperspectral remote sensing, which utilizes multiple narrow electromagnetic bands to obtain data about target objects, containing rich spatial, radiometric, and spectral information, has become a frontier field in current remote sensing technology. Unlike traditional RGB images, HSIs typically exist in the form of data cubes, containing not only rich spatial information but also spectral data spanning from tens to thousands of contiguous bands, endowing hyperspectral data with the characteristics and advantages of spectral continuity [8].

Among the various remote sensing tasks, HSI classification has always been a crucial field, referring to the pixel-level classification of HSIs using machine learning and deep learning methods on given hyperspectral data [9]. In supervised HSI classification, neural networks are typically trained using a subset of manually labeled data, allowing the model to learn the spatial and spectral characteristics of each land cover class. Subsequently, the trained model is used to predict unlabeled data [10], [11]. It is worth noting that the datasets used in the aforementioned classification tasks are usually collected by the same sensor, during the same time period, and under the same environmental conditions, meaning that these datasets adhere to the assumption of independent and identically distributed (i.i.d.) data. However, in practical applications, the data we often need to handle may not adhere to this assumption due to environmental or human factors, meaning that the same objects may exhibit different feature representations, which in turn can lead to the occurrence of domain shift problems in the HSI classification. The existence of domain shift problems makes the HSI classification more complex and challenging, requiring appropriate methods to address these issues.

Recently, the development of models, such as "domain adaptation" and "domain generalization (DG)" [40], has offered fresh perspectives on the cross-domain classification of HSIs. Hyperspectral cross-domain classification entails training a model on data from one source domain and subsequently applying the trained model to data from a different target domain. The goal is for the model to effectively handle domain shift and maintain strong classification performance (as illustrated in Fig. 1). Currently, there are two main approaches for hyperspectral cross-domain classification tasks: unsupervised domain adaptation (UDA) and single-source DG (SDG). The primary distinction between domain adaptation and DG is that UDA uses unlabeled or sparsely labeled data from the target

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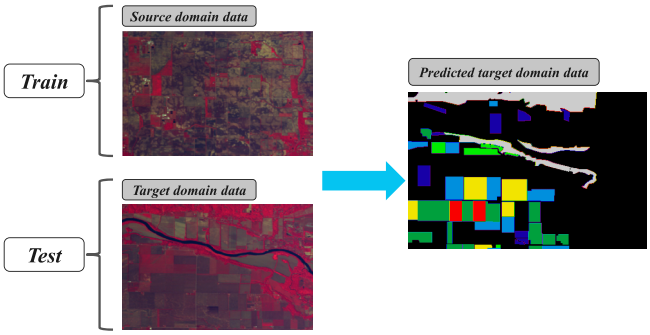


Fig. 1. Hyperspectral DG classification task.

domain during training, whereas SDG methods are trained entirely on the source domain without any prior knowledge of the target domain before testing. Collecting and annotating real-time data from the target domain poses significant challenges; therefore, DG tasks offer greater practical value compared to domain adaptation tasks [39].

Currently, mainstream domain adaptation methods mainly focus on feature alignment, which falls under the umbrella of homogeneous transfer learning [12], [13], [14], [15]. The basic concept is to identify shared invariant features between the target domain and the source domain to minimize domain shift between them. Inspired by generative adversarial network (GAN) frameworks, Ganin et al. [16] proposed domain adversarial neural networks (DANNs), where a generator is used for feature extraction, and a discriminator, combined with a classifier, learns domain-invariant representations. Yu et al. [21] leverage dynamic parameter learning to account for the different contributions of marginal and conditional distributions. In [17], graph neural networks are introduced, combining cross-domain few-shot learning with graph-based domain alignment to learn robust correspondences of nonlocal spatial information. Zhang et al. [18] employ graph optimal transport (GOT) combined with maximum mean discrepancy (MMD) to align the topological relationships, aiding in aligning the distributions between the source and target domains. Zhang et al. [19] propose discriminative coalignment of subspaces and distribution (DCA) to jointly reduce geometric and statistical shifts. Deng et al. [20] utilize metric learning and adversarial methods to encourage embedding features of source and target domain samples that are difficult to distinguish, thereby promoting clustering of target scene embeddings similar to those of the source scene.

Compared to domain adaptation, DG faces greater challenges as it lacks any prior knowledge of the target domain. Specifically, DG can be divided into multisource DG (MDG) and SDG. MDG generalizes effective information from multiple source domains, whereas SDG generalizes from a single source, making it more challenging. Typical approaches to DG involve domain augmentation and acquiring domain-invariant representations. Domain augmentation involves enriching and expanding the source domain data using various methods, with the requirement that the augmented data should be both related to and distinct from the source domain. The primary objective of domain-invariant representations is to extract data features that accurately represent the data and remain consistent across different domains [22], [23]. By employing a trainable spatial network module, the performance of the generator is significantly enhanced. Li et al. [38] propose a method for learning deep cross-modal embedding networks, which embed different modalities of data (such as images and text) into a common feature space, thus achieving zero-shot classification. Jiang et al. [39] introduce a remote sensing knowledge graph and construct a robust deep alignment network to enhance the model's generalization ability and recognition capability for new categories. Zhang et al. [24] employ a multimodal architecture with image and text encoders to extract visual and language features, utilizing coarse- and fine-grained text representations for language feature extraction at dual levels. In [25], the generator consists of a convolution-randomization-deconvolution structure, with morphological operation modules designed to extract domain-invariant information. Zhao et al. [26] utilize 3-D convolutions to simulate the generation domain, achieving slice-wise unbiased generators with high reliability and effectiveness in data generation. Dong et al. [27] employ the generator to enhance the source domain, while the discriminator uses a causal effect module to learn the causal relationship between features and outputs.

When facing the challenge of DG without prior knowledge of the target domain, the framework of GAN [1] provides an insightful approach to solving the DG problem. Currently, among adversarial methods, most approaches use randomization for data augmentation in the generator [25], [26], [27], which results in generated samples lacking effective modeling relationships and most randomization techniques lacking the powerful learning capability of convolutional neural networks (CNNs). To address this, we propose D3Net, where the generator is based on source domain data and integrates spatial feature learning modules with spectral information perturbation modules to generate data, thereby expanding the training samples. We contend that by learning domain-invariant features from the source domain, we can generate data with generalized features, enabling the trained model to adapt to relatively unfamiliar data for classification tasks. Compared to traditional simple feature perturbation or spatial transformations, a trained generative network module has been shown to better adapt to real-world domain shift phenomena [22]. In the discriminator, we introduce Vision Transformer (ViT) modules and convolutional modules for dual-feature extraction. We propose that different network modules contribute differently to category features. Therefore, we perform weighted selection on the logits output by the two network modules, allowing the network to progressively learn to express high-accuracy categories for each module. In addition, through contrastive learning and adversarial learning, the generator and discriminator iteratively update each other, thereby improving the overall performance of the model.

The primary contributions of this study can be outlined as follows.

1) We propose a trainable hyperspectral spatial information generation module that learns to identify domain-invariant information. Compared to traditional random perturbations at the feature level, it better simulates real-world domain shifts.

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2) For the first time, we integrate ViT into the discriminator module of the HSI DG model, leveraging its long-range dependencies to enhance the discriminator's feature discrimination capability.
3) We utilize a trainable tensor to weight the classification contributions of each module in the model, achieving comprehensive feature representation and enhancing classification confidence. Ablation studies validate the effectiveness of our approach.

II. Related Work

A. Domain Generalization

SDG involves training a model using data from only a single static domain, enabling it to extract domain-invariant information and aim for good performance in the target domain. Due to the limited information available from learning solely from a single-source domain, SDG faces greater challenges compared to MDG, making it more difficult to generalize the model to the target domain. Another similar task is UDA, both of which are mainstream methods for hyperspectral cross-domain classification. However, unlike UDA, SDG does not require unlabeled data from the target domain, thus demonstrating better real-time capabilities and generalization performance.

Learning domain-invariant representations is the most common approach for DG and is the goal of most DG models. Parascandolo et al. [30] discuss interpretable machine learning and propose a method to learn explanations that are difficult to change. Krueger et al. [31] propose a method to improve the generalization ability of deep learning models on unknown domain data through risk extrapolation. Lu et al. [32] present a technique for exploring domain-invariant features primarily for DG, aiming to improve the generalization performance of deep learning models on unfamiliar domains. Sagawa et al. [33] focus on the performance of distributionally robust neural networks when dealing with changes in data distribution, highlighting the significance of regularization for achieving robust generalization in worst case scenarios.

B. Contrastive Learning

A novel representation learning framework termed supervised contrastive learning is proposed [29], aiming to enhance the representation learning performance by combining supervised signals with contrastive learning. The core concept involves training the model by creating both positive and negative samples, ensuring that data features in the projection space are clustered closer within classes and farther apart between classes, thereby improving the accuracy of classification tasks. The correlation between positive and negative samples can be articulated as

$$S\left(F(x), F\left(x^{p}\right)\right) \gg S\left(F(x), F\left(x^{n}\right)\right) \tag{1}$$

where x, x^p, and x^n represent the original sample, positive sample, and negative sample, respectively; F is the feature extraction module; and $S(,)$ denotes the similarity between sample pairs, commonly based on Euclidean distance or cosine distance. The essence of this relationship lies in the notion that the similarity between pairs of positive samples should significantly outweigh that of negative sample pairs. Therefore, a simple contrastive loss can be designed as

$$\mathcal{L}_{\text{simple}}(x_i) = -S_{i,i} + \lambda \sum_{i \neq j} S_{i,j} \tag{2}$$

$$\frac{\partial \mathcal{L}_{\text{simple}}}{\partial S_{i,k}} = \lambda \tag{3}$$

where λ is a hyperparameter and $S_{i,j}$ represents the similarity between samples of classes i and j. Obviously, we aim for a larger value of $S_{i,i}$ and a smaller value of $S_{i,j}$.

However, the aforementioned contrastive learning method suffers from a common issue, namely, poor classification performance on hard samples (such as negative samples with very similar features). As shown in (3), from a gradient perspective, it is evident that the gradient of the simple contrastive loss for all samples is λ, which is obviously unreasonable for different positive and negative sample pairs. To achieve learning from contrastive samples as described above, the InfoNCE loss [34] is used to ensure the model correctly classifies positive and negative samples while facilitating self-discovery of hard samples. Its formula is represented as shown in the following equation:

$$\mathcal{L}_{\text{NCE}} = -\log \frac{\exp(S(F(x), F(x^{p}))/\tau)}{\sum_{j=1}^{N-1} \exp\left(S\left(F(x), F\left(x_j^{n}\right)\right)/\tau\right)} \tag{4}$$

$$\frac{\partial \mathcal{L}_{\text{NCE}}}{\partial S_{i,k}} = \frac{\exp\left(S_{i,j}/\tau\right)}{\tau\left(\sum_{k \neq i} \exp\left(S_{i,k}/\tau\right) + \exp\left(S_{i,i}/\tau\right)\right)} \tag{5}$$

where N represents the number of distinct categories and τ serves as a temperature hyperparameter, regulating the sensitivity of $S(\bullet)$. Presently, numerous experiments indicate that τ should be set to a relatively small value (we set it to 0.07), as it allows the model to more effectively distinguish challenging samples [35], [36]. The remaining parameters in the equation have the same meaning as in (2). Equation (5) clearly indicates that the InfoNCE contrastive loss will have the corresponding gradients based on different sample pairs, thereby facilitating the self-discovery of hard samples and enhancing the performance of the contrastive loss.

In our work, $\mathcal{L}_{\text{NCE}}$ is chosen as the contrastive loss function for all domains, aiming to enhance the model's classification performance through the construction of matching pairs and nonmatching pairs. In addition, to improve the effectiveness of the generated samples, an adversarial loss [25] based on this contrastive loss is introduced to distinguish between source data and generated data. The formulas for contrastive loss and adversarial loss presented here are as follows:

$$\mathcal{L}_{\text{con}} = -\sum_{i=0}^{N} \frac{1}{|P(i)|} \times \sum_{p \in P(i)} \log \frac{\exp\left(S\left(F(x), F\left(x_p^{+}\right)\right)/\tau\right)}{\sum_{n \in N(i)} \exp\left(S\left(F(x), F\left(x_n^{-}\right)\right)/\tau\right)} \tag{6}$$

$$\mathcal{L}_{\text{adv}} = -\sum_{c} \sum_{i=0}^{n_c} \frac{1}{|P_c(i)|} \times \sum_{p \in P_c(i)} \log \frac{\exp\left(S\left(F(x), F\left(x_p^{+}\right)\right)/\tau\right)}{\sum_{n \in N_c(i)} \exp\left(S\left(F(x), F\left(x_n^{-}\right)\right)/\tau\right)} \tag{7}$$

where c represents the number of distinct categories; n_c represents the total number of samples in class c; $P(i)$ and

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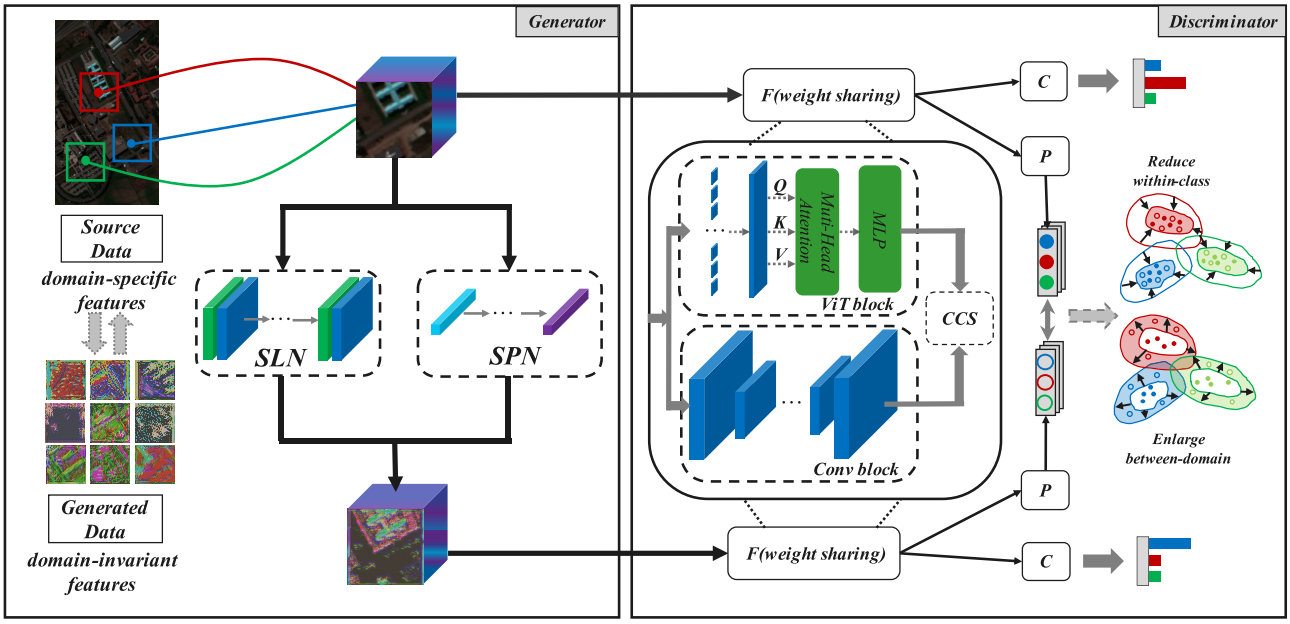


Fig. 2. Overall flowchart of D3net. Different circles represent different categories, with solid and hollow circles indicating sources from the source domain or the target domain, respectively. SLN: spatial learning module, SPN: spectral perturbation module, F: feature extraction module, CCS: classification contribution selection, and C and P: classification head and the projection head, respectively.

$N(i)$ represent the positive and negative sample sets, respectively; $P_c(i)$ and $N_c(i)$ represent the positive and negative sample sets of class c, respectively; and $|P(i)|$ and $|P_c(i)|$ represent the number of samples in the corresponding sets, respectively. In the contrastive loss, sample similarity based on class membership determines positive and negative samples, whereas in the adversarial loss, the sample domain (source or generated) determines positive and negative samples, with corresponding supervision labels set to 0 and 1, respectively. Viewed from a domain perspective, this approach fosters proximity between samples from the same domain in the projection space while promoting separation between samples from different domains.

C. Attention Mechanism

In [41], the attention mechanism was first proposed for processing textual information, laying the foundation for subsequent visual attention mechanisms. Dosovitskiy et al. [42] aim to demonstrate that the transformer model can perform as well as, or even surpass, traditional convolutional structures in visual tasks while keeping the overall framework of the transformer model unchanged. Traditional convolutional networks have inductive biases toward locality and translational invariance, which can limit the model's performance. However, these inductive biases are not present in the transformer architecture; instead, they are learned from a substantial volume of data. Consequently, the performance of transformer models, when trained on extensive datasets, can surpass that of traditional convolutional architectures. In addition, the ViT model incorporates a global attention mechanism, enabling it to establish long-range contextual dependencies between arbitrary image patches, an advantage not present in traditional convolutional structures.

Assume that the input data block size is $H \in R^{p^2 \times C}$, where p represents the side length of the image block (square) and C denotes the number of channels. First, tokenized $H \in R^{p^2 \times C}$ is divided into h equal parts, each denoted as $\hat{H}_\alpha \in R^{p^2 \times (C/h)}$, where h represents the number of attention heads and $\alpha = 1, 2, \ldots, h-1, h$. Then, $\hat{H}$ is linearly transformed to obtain Q (query), K (keyword), and V (value)

$$Q_\alpha \in R^{p^2 \times (C/h)} = W_{\text{linear_}q} \cdot \hat{H}_\alpha \tag{8}$$

$$K_\alpha \in R^{p^2 \times (C/h)} = W_{\text{linear_}k} \cdot \hat{H}_\alpha \tag{9}$$

$$V_\alpha \in R^{p^2 \times (C/h)} = W_{\text{linear_}v} \cdot \hat{H}_\alpha \tag{10}$$

where $W_{\text{linear_}*} \in R^{p^2 \times p^2}$ is the learnable matrix. Then, the multihead attention mechanism proceeds to compute the attention matrix for each head

$$A_\alpha = \text{Attention}(Q_\alpha, K_\alpha, V_\alpha) = \text{Softmax}\left(\frac{Q_\alpha \cdot K_\alpha^T}{\sqrt{d}}\right) \cdot V_\alpha \tag{11}$$

where $A_\alpha \in R^{p^2 \times (C/h)}$ and d equals the number of input channels in our work. Finally, the result of the multihead self-attention mechanism is expressed by the following equation:

$$\text{Output} \in R^{p^2 \times C} = \text{Concat}(A_1, A_2, \ldots, A_h) + H \tag{12}$$

where Concat() merges the attention matrices acquired from each head into a unified matrix.

III. Methodology

The network structure described in this article primarily consists of a generator and a discriminator. The generator is primarily composed of two parts: a spatial learning network (SLN) and a spectral perturbation network (SPN). The discriminator features a dual-feature embedding module, combined with classification and projection heads, as well as classification contribution weighted parameters. The adversarial nature of the model is primarily handled by the loss function. The overall model structure is illustrated in Fig. 2.

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The main process of the network is as follows. Specified patch-sized data blocks are extracted from the source domain and input into the generator. Within the generator, these data blocks pass through the spatial learning module and the spectral perturbation module and then go through decoding and fusion to generate new data. The structures of these two modules will be detailed later in this article. The generated data, along with the source data, are input into the discriminator, where the classification loss and adversarial loss between the two types of data are computed. The former is used to train the discriminator to correctly classify objects, while the latter ensures that the generated data deviates more from the source domain data. For detailed algorithms, refer to Algorithm 1.

Algorithm 1 Algorithm of D3Net

Input : Source domain samples $\mathbb{S} = \{x_i^s, y_i\}_{i=1}^{N_s}$, total epoch number T.
Output: The parameters $\theta_{SLN}, \theta_{SPN}$ of generator G, $\theta_F, \theta_C, \theta_P$ of discriminator D.

Initialize: $\theta_{SLN}, \theta_{SPN}, \theta_F, \theta_C, \theta_P$
for epoch $= 1$ **to** T **do:**
  Generate data from unknown domains in combination with the source domain
  $X_{SD} = \boldsymbol{Sampling}(\mathbb{S})$
  $X_{spa} = G_{SLN}(X_{SD}), X_{spe} = G_{SPN}(X_{SD})$
  $X_{GD} = \boldsymbol{Concat}\left(X_{spa}, X_{spe}\right)$
  Discriminator network for both source domain samples and generated samples:
  $Z_{SD} = D_C(X_{SD}), Z_{GD} = D_C(X_{GD})$
  $P_{SD} = D_P(X_{SD}), P_{GD} = D_P(X_{GD})$
  forall Z_{SD}, Z_{GD}:
    Calculate the loss $\mathcal{L}_{cls}$ through Eqs. (15)–(17)
    Calculate the loss $\mathcal{L}_{con}$ through Eq. (6)
    Calculate the D total loss $\mathcal{L}_D$ through Eq. (18)
  end
  forall P_{SD}, P_{GD}:
    Calculate the loss $\mathcal{L}_{adv}$ through Eq. (7)
    Calculate the G total loss $\mathcal{L}_G$ through Eq. (19)
  end
  Update $\theta_{SLN}, \theta_{SPN}$ by gradient descent
  Update $\theta_F, \theta_C, \theta_P$ by gradient descent
end

A. Domain-Invariant Feature Generator

1) Spatial Learning Network: In the spatial information learning module, we utilize an SLN to model the spatial characteristics of HSIs, aiming to enable the model to ignore domain-specific details and learn invariant spatial patterns across domains. The texture, shape, and logical spatial information (e.g., buildings are typically accompanied by roads nearby) of certain land cover types may remain relatively consistent across different datasets or environments, thereby being considered to possess domain invariance. Research by Zhou et al. [22] has shown that using a complete convolutional network to model domain transfer is more effective in addressing domain shift phenomena in real-world scenarios than traditional data augmentation techniques, such as simple spatial operations or feature

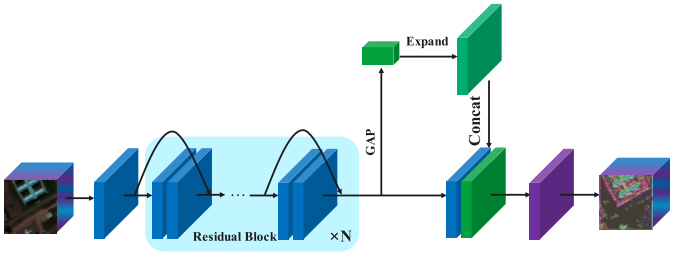


Fig. 3. SLN: aiming to learn domain-invariant spatial features.

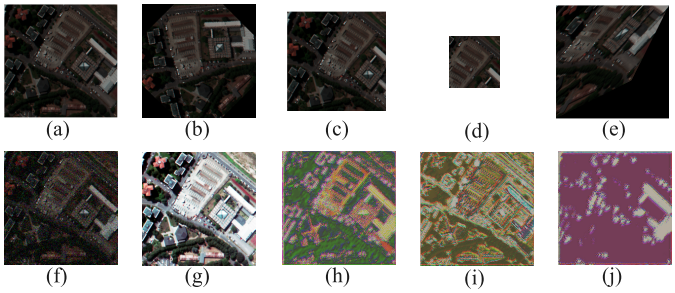


Fig. 4. Comparing different spatial enhancement methods. (a) Original image, (b) random flipping, (c) translation, (d) random cropping, (e) affine transformation, (f) adding noise, (g) contrast adjustment, and (h)–(j) images generated by SLN under different randomization parameters.

perturbations. The operations on spatial information are illustrated as follows:

$$\hat{x} = x + L_\theta(x). \tag{13}$$

Among them, L_θ is a complete learnable convolutional network.

Specifically, we start by extracting data blocks from the source domain, which are then input into the SLN. The data are initially converted into feature maps and then passed through N residual blocks (we set N to 4), where each residual block consists of a convolutional layer, an activation layer, and an instance normalization layer, cascaded twice. The output of the residual blocks undergoes global average pooling and resizing before being fused with the previous output. Through a series of network operations, domain-invariant spatial information is extracted from the source domain data. The data generated by the SLN effectively simulate domain shift phenomena observed in real-world scenarios. If the discriminator can classify these generated data well, it theoretically supports the hyperspectral cross-scene DG classification task. The detailed specifications of the SLN module are shown in Fig. 3.

In addition, Fig. 4 presents the visualizations of different data augmentation methods. Clearly, simple data augmentation operations and gradient-based simple perturbations only disturb features in the intermediate layers of the network. It is evident that Fig. 4(b)–(g) is essentially no different from Fig. 4(a), as it contains the same source domain-specific information and lacks adaptive learning capabilities. This clearly does not align with the domain shift phenomena observed in the real world. In contrast, as shown in Fig. 4(h)–(j), the SLN module can iteratively learn to capture high-level spatial semantic features, ignore domain-specific details, and achieve modeling of spatial domain invariance. The reasonable sample data generated by the network help the discriminator model learn domain-invariant features more effectively.

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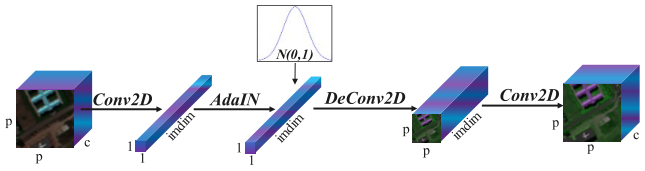


Fig. 5. SPN: aiming to enhance the robustness of spectral feature learning.

2) Spectral Perturbation Network: In HSIs, common confusing phenomena, such as "different spectra characteristics with the same object" and "different object with same spectrum," occur frequently. The complexity of spectra makes it very challenging to completely learn the spectral characteristics of various objects. Fortunately, in most cases, the spectral characteristics of objects within the same category are generally similar. Hence, we use the AdaIN [28] technique to introduce appropriate normal distribution perturbations to the spectral features. This helps the model adapt to spectral features under a broader range of errors, enhancing the robustness of the model's learning of spectral features, which positively impacts the model's DG performance.

The AdaIN is a normalization technology. Its original function is to transfer the style and texture in one picture (style map) to another picture (content map) while retaining the main structure of the content map. This article employs AdaIN to modify the mean and variance of the input data via learned parameters, thereby achieving stylized adjustments of the input data [43], [44]. The input data are adjusted through the learned parameters to adapt it to the given style vector, and its formula is shown in the following equation:

$$\mathrm{AdaIN}(z, n) = \mathrm{FC}_1(n)\frac{z - \mu(z)}{\sigma(z)} + \mathrm{FC}_2(n). \tag{14}$$

Among them, z is the content feature map, n is the transfer style, and FC_1 and FC_2 are both fully connected layers. In this article, $n \sim N(0, 1)$.

Within the network, we start by employing a convolutional kernel, sized equivalently to the patch dimensions, to transform the source domain data into spectral embedded features with dimensions of 1×1. Following this, we apply AdaIN techniques for perturbation before utilizing operations such as deconvolution to restore the image size, as shown in Fig. 5.

Overall, the generator has two main objectives. First, the generated data should be as different as possible from the source domain data to ensure that the generator ignores specific domain details. Second, the generated data should correctly complete the classification task for each category, ensuring that the generator does not alter the core features used for classification. This approach allows the generator to model the data reasonably well, leading to better generalization of the overall model.

B. Dual-Channel Feature Discriminator

1) Convolution Module: The convolutional module, with its advantages such as local connectivity, weight sharing, and hierarchical feature extraction, plays a crucial role in the field of computer vision. Therefore, we incorporate the convolutional module as part of our discriminator's feature extraction process. Specifically, our convolutional module receives data blocks and processes them through two cascaded convolution-ReLU activation-max-pooling modules, generating feature maps. These feature maps then pass through two fully connected FC-ReLU layers to obtain refined feature representations.

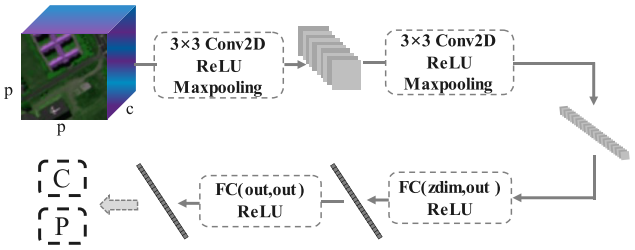


Fig. 6. Convolutional module.

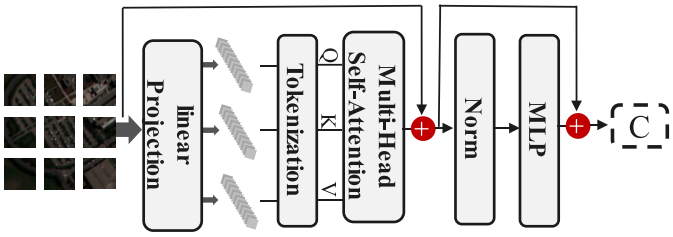


Fig. 7. ViT module.

Depending on the stage (training or testing), the obtained feature representations are connected to different output heads. During the training stage, both the classification head (C) and the projection head (P) are simultaneously connected. The classification head is used for training with labeled data, while the projection head provides high-dimensional projections for contrastive learning and adversarial learning. During the testing phase, only the classification head is connected to complete the classification task. The classification head is a linear layer, whereas the projection head is a linear layer combined with a ReLU activation function. The output dimensions of the linear layers in both heads are different. For details on the convolutional module, refer to Fig. 6.

2) ViT Module: While convolutional networks exhibit powerful feature extraction capabilities, they often lack global perception and struggle with long-range dependencies. These weaknesses happen to be strengths of the ViT model. Therefore, we choose ViT as another branch of the discriminator to complement the shortcomings of convolution. We made minimal modifications to the ViT model compared to [42]; we simply changed the input from image patches to individual pixels and adjusted the data input accordingly. In this article, the number of attention heads is set to 4, and the depth is set to 1. The ViT module is illustrated in Fig. 7.

3) Weighted Logits: ViT and convolution modules have their own distinct advantages, leading us to conclude that they contribute differently to the feature representations of various categories. In other words, convolution may excel in some categories, while ViT may outperform convolution in others. Therefore, we introduce a learnable 1-D tensor to weight the logits obtained from these two modules. Through iterative updates of the network, different weights are assigned to each module's contribution to the discrimination of different land objects.

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4) Loss Function: For the classification head, the classification loss helps to better distinguish between different types of land species, thereby improving the model's classification accuracy. For the projection head, contrastive loss brings similar categories closer together and pushes different categories apart, while adversarial loss makes data from the same source closer and separates generated data from the original data. Overall, this encourages the generated data to ignore source domain-specific details that are detrimental to generalization, thereby providing higher quality data for the discriminator.

In summary, this article employs classification loss, contrastive loss, and adversarial loss. For classification loss, traditional cross-entropy classification loss is used, as shown in formula (15). The classification losses for the source domain data and the generated data are represented by formulas (16) and (17), respectively. Therefore, the overall discriminator loss formula is given by formula (18)

$$\mathcal{L}_{\mathrm{CE}(p_i,y_i)} = -\sum_{c} y_i^c \log p_i^c \tag{15}$$

$$\mathcal{L}_{\mathrm{SD}(X_{\mathrm{SD}},Y)} = \frac{1}{N_{\mathrm{SD}}} \sum_{i} \mathcal{L}_{\mathrm{CE}(\mathrm{FC}(x_i^{\mathrm{SD}}),y_i)} \tag{16}$$

$$\mathcal{L}_{\mathrm{GD}(X_{\mathrm{GD}},Y)} = \frac{1}{N_{\mathrm{GD}}} \sum_{i} \mathcal{L}_{\mathrm{CE}(\mathrm{FC}(x_i^{\mathrm{GD}}),y_i)} \tag{17}$$

$$\min_{\theta_{\mathrm{FC}},\theta_P} \mathcal{L}_D = \mathcal{L}_{\mathrm{SD}} + \mathcal{L}_{\mathrm{GD}} + \lambda_1 \mathcal{L}_{\mathrm{con}}. \tag{18}$$

Among them, p_i is the predicted label, y_i is the real label, N is the number of samples, FC represents the feature embedding module with a classification head, and λ_1 is the hyperparameter used to control the contrastive loss.

To enhance the robustness of the model across different domains while maintaining classification performance, we use $\mathcal{L}_{\mathrm{adv}}$ and $\mathcal{L}_{\mathrm{GD}}$ as the loss functions for the generator. The former encourages the generated data to be distinct from the source domain data, while the latter promotes the correct classification of the generated data, as shown in the following equation:

$$\min_{\theta_G} \mathcal{L}_G = \mathcal{L}_{\mathrm{GD}} + \lambda_2 \mathcal{L}_{\mathrm{adv}} \tag{19}$$

where λ_2 is the hyperparameter used to control the adversarial loss. In this article, we set λ_1 equal to λ_2. $\mathcal{L}_{\mathrm{adv}}$ and $\mathcal{L}_{\mathrm{GD}}$ work together to ensure that the generated data deviate as much as possible from the original data without altering the core features used for classification.

IV. Experimental Section

Experiments are conducted on three datasets: Houston, Pavia, and Indiana. In the Houston dataset, the source domain is Houston13 and the target domain is Houston18. In the Pavia dataset, the source domain is PaviaU and the target domain is PaviaC. In the Indiana dataset, two locations are selected from the complete Indiana HSI [20], one as the source domain and one as the target domain (Fig. 8).

A. Datasets Used in Experiment

1) Houston Dataset: The Houston dataset contains Houston13 and Houston18, which were obtained by different sensors in the Houston University campus and its surrounding areas in 2013 and 2018, respectively. The Houston13 dataset image size is 349 × 1905 pixels and contains 144 spectral bands, the wavelength range is between 380 and 1050 nm, and its spatial resolution is 2.5 m. The wavelength range of the Houston18 dataset is between 380 and 1050 nm, including 48 spectral bands with a spatial resolution of 1 m. In order to achieve label unification between different domains, we selected the 209 × 955 pixel area where the two datasets overlapped and seven ground object categories common to both data. The categories and numbers of samples are listed in Table I, and their false-color maps and ground truth maps are shown in Fig. 9.



Fig. 8. Source of the cross-domain Indiana dataset.

TABLE I
SAMPLE POINT DETAILS OF HOUSTON CROSS-SCENE DATASET

Class		Number of Samples	
NO.	Name	Houston13 (Source)	Houston18 (Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential	319	5347
6	Non-residential	408	32459
7	Road	443	6365
Total		2530	53200

TABLE II
SAMPLE POINT DETAILS OF PAVIA CROSS-SCENE DATASET

Class		Number of Samples	
NO.	Name	PaviaU (Source)	PaviaC (Target)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584
Total		39332	39355

2) Pavia Dataset: The Pavia dataset includes Pavia University (PU) and Pavia Center (PC), which is collected using ROSIS sensors with a spectral coverage of 430–860 nm. The PU dataset size is 610 × 340 pixels and contains 103 spectral bands, and the spatial resolution is 1.3 m. The PC dataset size is 1096 × 715 pixels and contains 102 spectral bands. For the two datasets, this article uses their shared 102 spectral bands and seven common ground feature types. The categories and numbers of samples are listed in Table II, and their false-color maps and ground truth maps are shown in Fig. 10.

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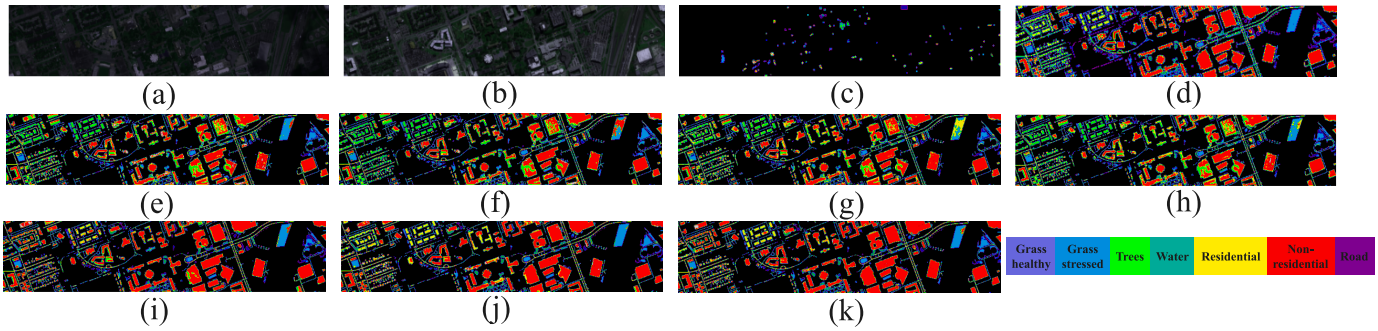


Fig. 9. Visualization of classification maps for different methods on the Houston18 dataset. (a) Visualization image of Houston 2013. (b) Visualization image of Houston 2018. (c) Ground truth label map of Houston 2013. (d) Ground truth label map of Houston 2018. (e) DIFEX (66.24%). (f) ANDMask (64.77%). (g) VREx (65.30%). (h) GroupDRO (67.47%). (i) SDEnet (77.68%). (j) LLURnet (75.37%). (k) D3net (81.02%).

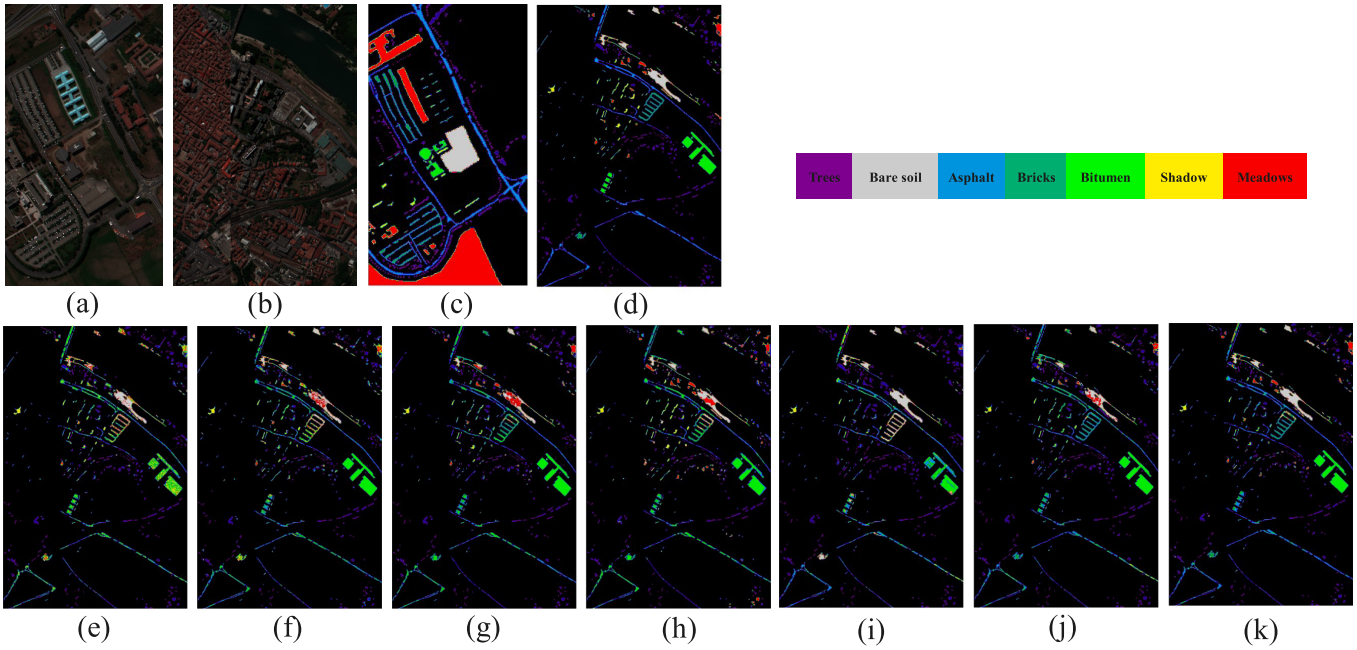


Fig. 10. Visualization of classification maps for different methods on the PaviaC dataset. (a) Visualization image of PaviaU. (b) Visualization image of PaviaC. (c) Ground truth label map of PaviaU. (d) Ground truth label map of PaviaC. (e) DIFEX (74.09%). (f) ANDMask (75.01%). (g) VREx (72.89%). (h) GroupDRO (71.75%). (i) SDEnet (80.96%). (j) LLURnet (81.77%). (k) D3net (85.18%).

3) Indiana Dataset: Indiana scene was acquired by the AVIRIS sensor in the Northwest Territories. Similar to the article [14], two independent subsets are selected as the source scene image and the target scene image (the two scenes have the same size, 400 × 300 × 220, and the number of spectral bands is 220). This article classifies a total of seven land object types in two scenes. The categories and numbers of samples are listed in Table III, and their false-color maps and ground truth maps are shown in Fig. 11.

TABLE III
SAMPLE POINT DETAILS OF INDIANA CROSS-SCENE DATASET

Class		Number of Samples	
NO.	Name	IndianaS (Source)	IndianaT (Target)
1	Concrete/Asphalt	4867	2942
2	Corn-CleanTill	9822	6029
3	Corn-CleanTill-EW	11414	7999
4	Soybeans-CleanTill	5106	1562
5	Soybeans-CleanTill-EW	4731	4792
6	Orchard	2996	1638
7	Wheat	3223	10739
Total		42159	35656

B. Parameter Tuning

The network model can be configured with numerous hyperparameters, each of which can have varying degrees of impact on the final model performance. Experiments compared the effects of hyperspectral data patch sizes {11, 13, 15, 17}, the number of residual blocks in the SLN {1, 2, 3, 4}, learning rates $\{1e^{-4}, 5e^{-4}, 1e^{-3}, 5e^{-3}\}$, and the contrastive adversarial loss function hyperparameter λ $\{1e^{-2}, 1e^{-1}, 1e^{-0}, 1e^{+1}\}$ on model performance. Fig. 12 depicts the overall accuracy (OA) performance.

C. Additional Details

Our experiments were conducted on a PyTorch framework utilizing a 13th Gen Intel Core i7-13620H processor and an NVIDIA GeForce RTX 4060 Laptop GPU with 8 GB of VRAM. We employed the Adam optimizer for our experiments. The learning rate for all datasets is set to 0.001, with a momentum of 0.9 and a sliding window stride of 1. For all cross-scene datasets, our aim is to establish correspondence between labels across different domains, thus only selecting

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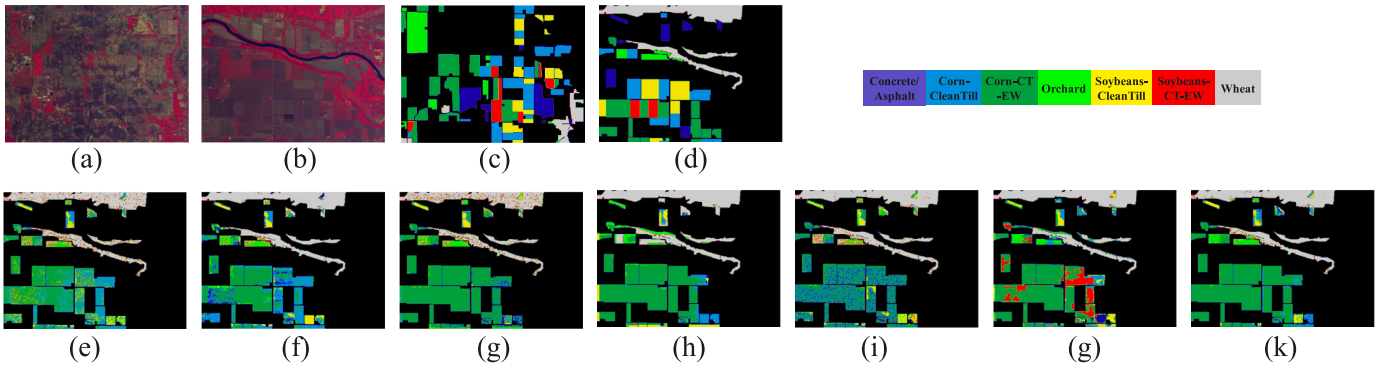


Fig. 11. Visualization of classification maps for different methods on the IndianaT dataset. (a) Visualization image of IndianaS. (b) Visualization image of IndianaT. (c) Ground truth label map of IndianaS. (d) Ground truth label map of IndianaT. (e) DIFEX (54.30%). (f) ANDMask (54.70%). (g) VREx (54.43%). (h) GroupDRO (54.23%). (i) SDEnet (53.65%). (j) LLURnet (50.78%). (k) D3net (55.4%).

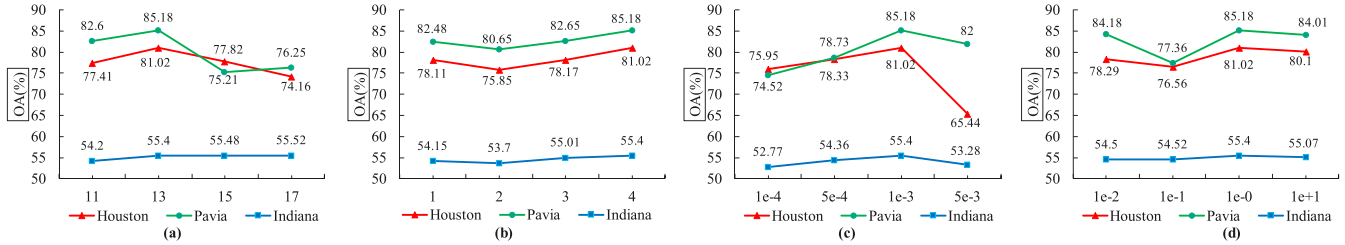


Fig. 12. Comparisons of performance (OA%) with different hyperparameters. (a) Patch size. (b) Resdual block. (c) Learning rate. (d) λ.

the common land cover classes between the two datasets. For the Houston13, PaviaU, and IndianaS datasets, 80% of the source domain data are used for training, 20% for validation, and all target domain data are used for testing. Notably, due to the relatively small number of training samples in the Houston dataset, we performed a fivefold data augmentation. We use OA, average accuracy (AA), and Kappa coefficient as performance metrics. We conducted five experiments using different random seeds and calculated the mean and standard deviation of the results to reduce the randomness of the experiments.

D. Experiment Results

To test the effectiveness of our model, we selected several state-of-the-art network models for comparison experiments. DIFEX [32] utilizes a knowledge distillation framework to capture advanced Fourier phases as internal invariant features and learn cross-domain correlation alignment as mutually invariant features. ANDMask [30] is aimed at enhancing the interpretability of the model's input data while preventing overfitting and memorization. The VREx [31] paper introduces a method for achieving generalization to unknown distributions through risk extrapolation, which focuses on generalizing to situations beyond the training data distribution. GroupDRO [33] explores how to design and train distributionally robust neural networks when facing group shifts, emphasizing the importance of regularization in enhancing model robustness. LLURnet [25] enhances the model's ability to learn spectral features by introducing local linear transformations to the data during training while preserving the original data distribution, thereby increasing data diversity. SDEnet [26] enhances data diversity and aids the model in better learning spectral features by introducing a domain expansion mechanism to generate a virtual domain from a single data source. They are all advanced methods for the DG task. For all experiments, we saved the best model performance. In addition, to illustrate the domain shift phenomenon in cross-domain hyperspectral datasets, we trained a 2D-CNN network on the source domain and directly tested it on the target domain [source only (SO)]. To demonstrate the representation learning capability of DG methods, we directly trained and tested the 2D-CNN on the target domain [train on target (TOT)], and 10% of the data were used for training and 90% for testing. In this scenario, the SO scene serves as the lower bound of model performance, while the TOT scene serves as an upper bound to some extent. These two baselines underwent a comparative experiment of 100 epochs each.

Tables IV–VI present the experimental results, while Figs. 9–11 illustrate the classification performance. It is important to note that due to differences in features such as sample size, spectral bands, and spatial properties among different datasets, the impact of the model can vary significantly. For instance, in the Houston dataset, the Houston13 source domain dataset contains only 2530 samples, whereas Houston18 has 53 200 samples. In contrast, the sample sizes of the source and target domains are very close in the Pavia dataset, resulting in relatively better performance across all models compared to other datasets. However, in the Indiana dataset, despite sufficient sample sizes for both domains, the high similarity among different features poses challenges for the model's recognition and feature learning, leading to inferior performance compared to other datasets.

A notable phenomenon is that in the Houston dataset, the SO method achieves better OA compared to some DG methods. This is easily explained by the fact that the Houston13 and Houston18 images are taken entirely from the same spatial region, with most of the features overlapping. The 2D-CNN can fully utilize these common feature characteristics for learning. When the model is applied to the test set Houston18, it performs relatively well on this cross-domain dataset due to the ample prior knowledge. However, this advantage does not exist in other datasets, leading to a significant performance

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TABLE IV
CLASSIFICATION RESULTS OF OUR METHOD AND COMPARATIVE EXPERIMENTS ON HOUSTON

Houston datasets										
No.	name	SO(CNN)	TOT(CNN)	DIFEX[32]	ANDMask[30]	VREx[31]	GroupDRO[33]	LLURnet[26]	SDEnet[25]	D3Net(Ours)
1	Grass healthy	40.95	73.78	52.75	54.89	55.12	45.08	46.61	77.01	46.68
2	Grass stressed	77.79	96.14	60.99	54.84	59.90	63.96	87.17	69.70	81.81
3	Trees	51.98	92.79	50.15	53.89	50.74	55.90	68.00	59.04	58.60
4	Water	96.36	65.00	17.41	15.16	14.62	29.45	100	100	100
5	Residential	71.73	94.29	43.60	40.87	40.17	48.00	71.98	54.20	71.49
6	Non-residential	85.80	98.85	91.16	90.68	91.82	86.06	78.27	88.59	88.22
7	Road	19.33	93.43	34.70	36.67	33.99	41.25	64.27	56.06	53.39
OA(%)		72.80±1.66	96.53±0.15	65.38±2.27	64.91±2.68	63.82±2.05	65.28±3.73	76.63±3.33	78.07±1.61	**79.20±0.40**
AA(%)		63.42±9.98	87.75±3.54	50.11±5.05	49.57±2.24	49.48±3.67	52.81±6.69	**73.28±4.81**	72.39±5.08	71.51±2.51
Kappa×100		51.72±6.44	94.10±0.25	51.47±6.80	47.51±2.57	46.81±2.84	48.08±4.02	61.35±1.05	61.72±2.54	**64.60±1.03**

TABLE V
CLASSIFICATION RESULTS OF OUR METHOD AND COMPARATIVE EXPERIMENTS ON PAVIA

Pavia datasets										
No.	name	SO(CNN)	TOT(CNN)	DIFEX[32]	ANDMask[30]	VREx[31]	GroupDRO[33]	LLURnet[26]	SDEnet[25]	D3Net(Ours)
1	Tree	93.65	98.93	84.60	81.51	80.57	80.30	81.29	83.58	90.62
2	Asphalt	86.63	99.58	77.07	67.21	74.10	75.23	84.80	86.82	86.59
3	Brick	60.19	98.69	38.92	22.67	30.30	36.09	65.78	59.42	60.10
4	Bitumen	0.79	99.50	82.23	85.06	82.19	82.55	86.5	80.93	82.84
5	Shadow	79.62	99.98	91.59	86.79	88.59	87.02	86.44	85.87	88.95
6	Meadow	70.70	95.43	58.49	54.63	60.40	54.96	79.88	78.65	74.69
7	Bare soil	79.43	99.97	82.03	82.60	84.07	82.61	81.50	78.54	80.94
OA(%)		67.32±1.11	99.15±0.25	75.28±4.01	73.10±2.48	75.52±3.31	74.91±3.10	82.32±0.91	81.14±2.19	**83.16±1.19**
AA(%)		67.29±2.39	98.87±0.47	73.56±3.07	68.64±2.21	71.46±3.31	71.25±2.57	**80.88±1.57**	79.11±2.76	80.67±2.75
Kappa×100		60.81±1.64	98.97±0.31	70.32±4.88	67.35±2.93	66.94±4.07	69.62±3.76	78.78±1.08	77.32±2.60	**79.71±1.49**

TABLE VI
CLASSIFICATION RESULTS OF OUR METHOD AND COMPARATIVE EXPERIMENTS ON INDIANA

Indiana datasets										
No.	name	SO(CNN)	TOT(CNN)	DIFEX[32]	ANDMask[30]	VREx[31]	GroupDRO[33]	LLURnet[26]	SDEnet[25]	D3Net(Ours)
1	C/A	62.16	54.49	0.00	28.65	34.35	42.09	11.52	5.82	2.00
2	Corn-CleanTill	0.03	60.19	31.34	17.40	23.75	22.32	9.13	14.38	14.5
3	Corn-CT-EW	10.70	71.06	34.76	35.77	35.42	34.64	73.56	79.61	84.77
4	Soybeans-CT	83.75	92.97	70.02	77.42	72.30	70.74	87.00	88.82	88.37
5	S-CT-EW	4.47	82.42	0.00	0.06	0.00	0.00	0.35	0.05	0.01
6	Orchard	0.00	33.57	0.00	0.00	0.00	0.00	0.10	0.00	0.00
7	Wheat	95.11	99.41	92.74	96.28	96.34	96.78	95.75	96.17	97.76
OA(%)		40.91±2.95	77.15±7.37	54.27±0.96	54.30±0.67	54.29±0.42	54.17±0.63	51.64±1.57	53.81±0.75	**54.88±0.54**
AA(%)		37.49±1.46	70.59±7.87	37.58±4.93	36.51±1.97	37.45±2.36	38.09±3.63	39.64±0.68	40.81±0.49	**41.06±0.66**
Kappa×100		30.40±2.23	71.29±9.43	41.04±1.14	40.86±0.76	40.83±0.67	40.63±0.89	38.18±1.31	40.30±0.82	**41.31±0.64**

TABLE VII
MODEL COMPLEXITY OF DIFFERENT METHODS ON THE THREE DATASETS (SECONDS)

No.	Methods	DIFEX [32]	ANDMask [30]	VREx [31]	GroupDRO [33]	SDEnet [26]	LLURnet [25]	D3net
Houston	train	617.93	11.57	8.31	7.82	9.29	14.06	14.34
	test	13.49	10.46	10.33	10.75	7.52	11.73	10.20
Pavia	train	799.84	12.57	9.04	8.6	30.54	37.15	32.80
	test	13.83	13.03	12.94	13.75	11.55	12.38	12.09
Indiana	train	1094.47	13.99	10.64	9.96	43.39	52.5	41.94
	test	16.36	23.16	21.94	22.7	21.96	21.88	23.33

drop for the source-only method due to domain shift issues. In contrast, the TOT method, as a noncross-domain approach, operates under the assumption that the training and test sets are i.i.d., hence its exceptional performance across all datasets. To some extent, it can be considered the upper limit of cross-domain classification performance.

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TABLE VIII
PERFORMANCE IMPACT OF DIFFERENT MODULES ON D3NET

NO.	Class	No-SLN	No-SPN	No-Adv	No-Con	No-ViT	No-Conv	D3Net
1	Tree	93.25	89.25	90.43	84.23	84.14	**96.62**	81.25
2	Asphalt	89.44	87.04	**91.05**	90.71	88.86	45.11	88.99
3	Brick	19.33	86.74	51.77	**88.75**	87.6	58.96	85.25
4	Bitumen	80.25	70.22	79.09	77.69	80.47	75.19	**86.48**
5	Shadow	68.32	80.06	88.58	94.06	80.23	**94.31**	89.98
6	Meadow	66.25	71.52	66.89	76.28	77.86	55.86	**82.65**
7	Bare soil	86.76	77.13	**87.65**	72.24	78.14	69.55	81.97
OA(%)		79.88	80.95	83.39	82.93	83.03	70.08	**85.18**
AA(%)		71.94	80.28	79.35	83.42	82.47	70.8	**85.22**
Kappa×100		75.45	77.09	79.92	79.57	79.66	64.11	**82.24**

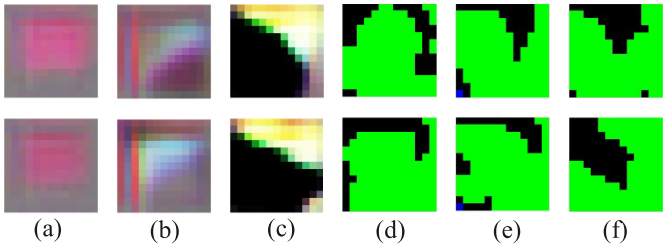


Fig. 13. Display of original and generated images at different training stages. The top row shows source domain data, and the bottom row shows generated data. (a)–(c) Early training phase. (d)–(f) After 20 epochs.

Based on the experimental results, it is evident that our model achieved the best OA and Kappa scores on the Houston dataset, Pavia dataset, and Indiana dataset, showing advantages over other models. Specifically, on the Houston dataset, our model's OA outperformed all existing state-of-the-art methods by 1.13–15.38 percentage points, and the Kappa coefficient also significantly exceeded others, with AA slightly lower than SDEnet and LLURnet. On the Pavia dataset, our model's OA led by 0.84–10.06 percentage points, and Kappa coefficients comprehensively outperformed other models. On the Indiana dataset, while the performance differences of most models were not significant, our model still maintained a certain advantage in OA by 0.58–3.24 percentage points. Table VII shows the time required to train one epoch and test different methods on the same device, reflecting the time performance of the different methods. In our approach, the use of CNNs in the generator instead of conventional randomization, along with the ViT encoding structure in the discriminator, results in a moderate increase in time consumption. However, experiments demonstrate that these structures significantly enhance the classification performance, and thus, we consider the tradeoff of slightly increased time consumption for improved classification performance to be acceptable.

Overall, our model demonstrates the best overall performance across the three experimental datasets, which proves that our model has strong DG capabilities. In addition, our model's leading performance on the Houston dataset indicates that it can achieve excellent generalization even with relatively insufficient sample sizes.

E. Ablation Study

To validate the efficacy of each module in the D3net network model, we conducted ablation experiments on the Pavia dataset used in our study. These experiments evaluated the contributions of the generator module and discriminator module to the overall performance. Specifically, we assessed the impact of SPN, SLN, contrastive loss, adversarial loss, convolutional module, and ViT on the model's performance. The experimental results are presented in Table VIII.

From the data in the table, it is evident that all the proposed modules positively impact the model's performance. Specifically, the SLN and Conv modules have a significant influence on overall performance, as their removal leads to a substantial decrease in accuracy. Fig. 13 shows the visualization effects of the data generated by the SLN module at different training stages. It is clear that in the initial stages of training, the SLN module merely mimics the original data. However, after further training, the module can accurately capture the primary spatial information and reasonably extend it to gradually form data that aligns with real-world domain shifts. Moreover, the classification accuracy of each category in the "No-ViT" and "No-Conv" settings demonstrates the varying contributions of different modules to specific category classifications. This indicates that the classification contribution parameters positively affect discrimination. Overall, the ablation experiments validate the rationality of our network design.

TABLE IX
CLASSIFICATION COMPARISON RESULTS BETWEEN METHODS WITH AND WITHOUT ID

	Compare	OA	AA	Kappa
Houston	Ours	**81.02**	**71.87**	**66.9**
	Ours+ID	60.80	65.02	44.62
Pavia	Ours	**85.18**	**85.22**	**82.24**
	Ours+ID	76.98	74.68	72.07
Indiana	Ours	**55.4**	**41.59**	**41.93**
	Ours+ID	54.35	40.05	40.52

F. Supplementary Experiments

1) Intermediate Domain: In some generative adversarial-based HSI DG methods, randomization parameters are used to weight the source domain and the generated domain to obtain an intermediate domain (ID) [25], [26], [27]. This approach aims to facilitate the transition between the source domain and the generated domain in cases of significant domain shift, thereby reducing the model's learning complexity. However, experimental comparisons reveal that adding the ID did not improve the learning capability of our model. We infer that the ID generated through random weighting may have introduced confusing features, thereby reducing the model's classification performance. The experimental results across the three datasets are presented in Table IX.

2) Swap Source Domain and Target Domain: In all cross-domain classification tasks involving hyperspectral data, the relationship between the source and target domain datasets is always fixed. However, as a DG task, the order of the source and target domains should be irrelevant. Therefore, as a supplementary experiment, we swapped the roles of the source and target domains in the datasets used in the experiments. Combined with an analysis of the experimental sample data,

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TABLE X
CLASSIFICATION RESULTS AFTER SWAPPING THE SOURCE AND TARGET DOMAINS

	Compare	OA	AA	Kappa
Houston	Ours	81.02	71.87	66.9
	Ours-EXC	**82.21**	**80.52**	**79.08**
Pavia	Ours	**85.18**	**85.22**	**82.24**
	Ours-EXC	66.14	69.09	55.92
Indiana	Ours	**55.4**	**41.59**	**41.93**
	Ours-EXC	39.69	35.94	22.87

this further demonstrates the generalization performance of our model. The experimental results are presented in Table X.

Clearly, in the Houston dataset, where the Houston18 dataset has a sample count of 53 200, swapping the roles of the source and target domains leads to more comprehensive model training. Consequently, there is a significant improvement in various metrics (even though we did not perform random flipping and random radiometric noise augmentation on Houston18), indicating that the types and quantities of samples in the source domain indeed have a crucial impact on model training. In the Pavia and Indiana datasets, although there is not much difference in the total sample count between the source and target domains, it is evident that the distribution of sample points for various land cover classes in the source domain is much better than that in the target domain. Therefore, the model can effectively learn the features of each land cover class, resulting in superior performance on the dataset before the domain swap.

V. CONCLUSION

This article proposes a network model based on a generative adversarial framework for HSI DG classification tasks. Specifically, the generator in the network aims to extract the core features used for classification from the source domain data and passes them to the discriminator in the form of generated data. This enables the discriminator to learn domain-invariant information, allowing the model to achieve good classification performance in the target domain without any prior knowledge. We conducted a series of experiments on several public datasets, and the results demonstrate the effectiveness of our method.

However, there are still many unexplored methods and research directions. For instance, the generator in this article mainly focuses on learning spatial domain-invariant information. However, it remains unknown whether the rich and complex spectral information also contains domain-invariant information. Exploring this aspect could be a crucial direction for future research. In addition, since current cross-domain tasks often exhibit some degree of correlation between the source and target domains, exploring the potential for achieving complete DG could be another important direction for future research.

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